

Question: 1

Given a company with 3 departments (HR: 20 users, Sales: 50 users, Tech: 100 users), assign appropriate subnets using VLSM from the 192.168.10.0/24 network and write the subnet address, subnet mask, and usable IP range for each department?

Answer: 1. Tech Department (Requires 100 users)

Subnet Size Needed: Block of 128 IPs ($2^7 = 128$), gives 126 usable addresses).

Subnet Address: 192.168.10.0/25

Subnet Mask: 255.255.255.128

Usable IP Range: 192.168.10.1 to 192.168.10.126

2. Sales Department (Requires 50 users)

Subnet Size Needed: Block of 64 IPs ($2^6 = 64$), gives 62 usable addresses).

Subnet Address: 192.168.10.128/26

Subnet Mask: 255.255.255.192

Usable IP Range: 192.168.10.129 to 192.168.10.190

3. HR Department (Requires 20 users)

Subnet Size Needed: Block of 32 IPs ($2^5 = 32$), gives 30 usable addresses).

Subnet Address: 192.168.10.192/27

Subnet Mask: 255.255.255.224

Usable IP Range: 192.168.10.193 to 192.168.10.222

Question: 2

You are given three office locations (Ahmedabad, Surat, Rajkot). Each office has three departments: Admin (10 users), Marketing (30 users), and Support (60

users). Design an IP addressing scheme using VLSM for all offices using the 10.0.0.0/22 block, ensuring no overlap and efficient usage. List each subnet with its office, department, subnet mask, and first/last usable IP?

Answer: 1. Support Departments (60 users $\rightarrow$) Block of 64 IPs / Mask: 255.255.255.192 or /26)

Ahmedabad (Support): Subnet: 10.0.0.0/26 | First Usable: 10.0.0.1 | Last Usable: 10.0.0.62

Surat (Support): Subnet: 10.0.0.64/26 | First Usable: 10.0.0.65 | Last Usable: 10.0.0.126

Rajkot (Support): Subnet: 10.0.0.128/26 | First Usable: 10.0.0.129 | Last Usable: 10.0.0.190

2. Marketing Departments (30 users $\rightarrow$) Block of 32 IPs / Mask: 255.255.255.224 or /27)

Ahmedabad (Marketing): Subnet: 10.0.0.192/27 | First Usable: 10.0.0.193 | Last Usable: 10.0.0.222

Surat (Marketing): Subnet: 10.0.0.224/27 | First Usable: 10.0.0.225 | Last Usable: 10.0.0.254

Rajkot (Marketing): Subnet: 10.0.1.0/27 | First Usable: 10.0.1.1 | Last Usable: 10.0.1.30

3. Admin Departments (10 users $\rightarrow$) Block of 16 IPs / Mask: 255.255.255.240 or /28)

Ahmedabad (Admin): Subnet: 10.0.1.32/28 | First Usable: 10.0.1.33 | Last Usable: 10.0.1.46

Surat (Admin): Subnet: 10.0.1.48/28 | First Usable: 10.0.1.49 | Last Usable: 10.0.1.62

Rajkot (Admin): Subnet: 10.0.1.64/28 | First Usable: 10.0.1.65 | Last Usable: 10.0.1.78

Question: 3

A startup has offices in Mumbai, Pune, and Baroda. Each office will connect to HQ via VPN. Propose a route summarization for all three offices if their assigned subnets are: Mumbai (172.16.10.0/24), Pune (172.16.11.0/24), Baroda (172.16.12.0/24). Write the summarized route and explain why it is efficient.

Hint: Find the smallest supernet that includes all three subnets.

Answer: 1. The Summarized Route (Supernet)

Summary Address: 172.16.8.0/22

Subnet Mask: 255.255.255.0 to 255.255.252.0

2. How the Summarized Route is Calculated (Binary Matching)

To find the smallest supernet, we look at the third octet of each network address in binary form:

Mumbai (10): 000010 10

Pune (11): 000010 11

Baroda (12): 000011 00

Matching Bits: The first 22 bits of all three networks are completely identical (172.16.000010xx.xxxxxxxx). The shared match ends at the 22nd bit position, creating a /22 summary boundary starting at decimal block 8.

3. Why It Is Efficient

Smaller Routing Tables: Instead of the main corporate HQ core router maintaining 3 individual route entries (one for Mumbai, Pune, and Baroda), it only has to manage 1 single summary entry (172.16.8.0/22) for all incoming VPN branch traffic.

Reduces CPU Overhead: Core routers process network traffic much faster because they spend less time searching through massive internal route lookups.

Isolates Network Flaps: If the internet connection at the Baroda office goes down temporarily, the local connection drop does not force the HQ core router to recalculate its entire network topology, keeping the primary network highly stable.

Question: 4

Given a scenario where the Tech department in Ahmedabad grows from 100 to 200 users, update your previous VLSM plan to accommodate the new size with minimal changes to the rest of the network. List the new subnet details and explain your approach.

Constraint: Do not change the existing subnets for HR or Sales unless absolutely necessary?

Answer: 1. Updated Ahmedabad Subnet Details

Tech Department (New Subnet): 192.168.10.0/24

Subnet Mask: 255.255.255.0

Usable IP Range: 192.168.10.1 to 192.168.10.254

Sales Department (Unchanged): 192.168.10.128/26 (Usable: .129 to .190)

HR Department (Unchanged): 192.168.10.192/27 (Usable: .193 to .222)

2. The Minimal Change Approach (Overlapping Exceptions)

No Changes to HR and Sales: In a traditional network redesign, changing Tech to a /24 would overwrite the .128 and .192 spaces used by Sales and HR. However, under the constraint to leave them untouched, we apply Intra-subnetting/VLAN segmentation.

Router-on-a-Stick / Layer 3 Switching: By keeping Sales and HR exactly where they are, the router can treat the entire 192.168.10.0/24 space as the Tech VLAN, but explicitly exclude the specific IP sub-pools of Sales and HR from the Tech DHCP pool.

Alternative (Separate Expansion Block): If physical address overlapping is entirely forbidden by hardware, a secondary, non-contiguous block (such as 192.168.11.0/25) must be routed to Tech to handle the extra 100 users, keeping the original 192.168.10.0/25 completely intact.

Question: 5

Create 5 VLSM subnetting challenge questions inspired by real apps you use (e.g., Zomato delivery zones, Flipkart warehouse locations, or IPL team fan clubs). For each, provide the required number of hosts and a base network address?

Answer: 1. Zomato Ahmedabad Delivery Zones

Scenario: Zomato wants to allocate unique subnets to three major delivery zones in Ahmedabad based on daily order volumes. The zones require different capacities to accommodate riders and tracking devices.

Base Network Address: 192.168.50.0/24

Host Requirements:

Satellite Zone (Support): 80 hosts

Bodakdev Zone (Tech): 45 hosts

Vastrapur Zone (Admin): 22 hosts

2. Flipkart Fulfillment Center Inventory

Scenario: A Flipkart automated warehouse needs to segregate its internal devices into three secure operational subnets to prevent scanner traffic from slowing down backend inventory syncing.

Base Network Address: 172.16.100.0/24

Host Requirements:

IoT Conveyor Belt Sensors: 115 hosts

Handheld Barcode Scanners: 55 hosts

Admin & Billing Terminals: 12 hosts

3. IPL Ahmedabad Stadium Fan Zones

Scenario: During the IPL season at the Narendra Modi Stadium in Ahmedabad, the franchise sets up temporary local Wi-Fi infrastructure for three dedicated fan experience zones.

Base Network Address: 10.20.0.0/22

Host Requirements:

Premium Corporate Box Lounge: 480 hosts

General Fan Club Zone: 220 hosts

Media & Press Box Enclosure: 90 hosts

4. Swiggy Instamart Dark Stores

Scenario: Swiggy Instamart is expanding its hyper-local grocery network. The central network team must allocate IP blocks for a cluster of three micro-fulfillment centers (dark stores) with varying staff sizes.

Base Network Address: 192.168.1.0/24

Host Requirements:

Dark Store A (Large Hub): 60 hosts

Dark Store B (Medium Hub): 28 hosts

Dark Store C (Small Hub): 14 hosts

5. Jio Cinema Streaming Server Cluster

Scenario: Jio Cinema needs to allocate local management subnets for its regional content delivery network (CDN) edge nodes to handle live sports traffic efficiently.

Base Network Address: 10.100.4.0/23

Host Requirements:

Video Cache Nodes: 240 hosts

User Authentication Nodes: 110 hosts

Monitoring Terminals: 30 hosts